REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

Generated: 2026-09-16 20:31:17
Reporting Period: All Recorded Traffic

1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	14736
Normal Flows	13725
Detected Attacks	1011
Attack Rate	6.86%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.50
Validation Accuracy	0.76%
Attack Precision	0.96
Attack Recall	0.60
Attack F1-Score	0.74

3. Network Protocol Distribution

Protocol	Flow Count
HTTPS	4713
TCP	4040
UDP	3392
DNS	1356
QUIC_OR_UDP443	1132
HTTP	53
IPSEC	32
DHCP	14
OTHER	2
ICMP	1
NTP	1

4. Detected Attack History

Time	Source	Destination	Protocol	Threat Vector	Severity	Confidence
2026-09-14 18:29:28	192.168.1.10:64764	3.6.199.65:443	HTTPS	ANOMALOUS FLOW	HIGH	90.0%
2026-09-14 18:28:57	104.18.39.21:443	192.168.1.10:57752	HTTPS	ANOMALOUS FLOW	HIGH	94.0%
2026-09-14 18:28:54	3.6.199.65:443	192.168.1.10:64289	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 18:28:50	192.168.1.10:52100	35.186.224.35:443	HTTPS	ANOMALOUS FLOW	HIGH	98.0%
2026-09-14 18:28:47	3.6.199.65:443	192.168.1.10:64764	HTTPS	ANOMALOUS FLOW	HIGH	98.0%
2026-09-14 18:28:47	192.168.1.10:57621	192.168.1.255:57621	UDP	ANOMALOUS FLOW	HIGH	94.0%
2026-09-14 18:28:45	13.202.220.237:443	192.168.1.10:59054	HTTPS	ANOMALOUS FLOW	HIGH	74.0%
2026-09-14 18:28:44	3.6.199.65:443	192.168.1.10:52205	HTTPS	ANOMALOUS FLOW	HIGH	74.0%
2026-09-14 18:28:43	8.8.8.8:53	192.168.1.10:62606	DNS	ANOMALOUS FLOW	HIGH	84.97%
2026-09-14 18:28:42	192.168.1.10:62803	142.251.151.119:443	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	90.0%
2026-09-14 18:28:41	104.18.39.21:443	192.168.1.10:57752	HTTPS	ANOMALOUS FLOW	HIGH	90.0%
2026-09-14 18:28:25	13.202.220.237:443	192.168.1.10:59054	HTTPS	ANOMALOUS FLOW	HIGH	70.0%
2026-09-14 18:28:24	3.6.199.65:443	192.168.1.10:52205	HTTPS	ANOMALOUS FLOW	HIGH	70.0%
2026-09-14 18:25:50	192.168.1.10:61935	52.123.253.91:443	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 18:25:50	192.168.1.10:52100	35.186.224.35:443	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 18:25:50	69.41.167.90:443	192.168.1.10:55179	HTTPS	ANOMALOUS FLOW	HIGH	56.27%
2026-09-14 18:25:46	192.168.1.10:57621	192.168.1.255:57621	UDP	ANOMALOUS FLOW	HIGH	95.0%
2026-09-14 18:25:45	8.8.8.8:53	192.168.1.10:56220	DNS	ANOMALOUS FLOW	HIGH	84.67%
2026-09-14 18:25:44	192.168.1.10:55245	10.2.0.1:65432	TCP	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 18:25:42	104.18.39.21:443	192.168.1.10:57752	HTTPS	ANOMALOUS FLOW	HIGH	100.0%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.